

## TULUN, Russian Federation

WMO: 305040

Lat: 54.562N

Lon: 100.579E

Elev: 1714

StdP: 13.81

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -32.5      | -26.7 | -36.6                         | 0.7 | -32.3 | -31.0 | 1.0 | -26.3 | 13.9                  | 8.0  | 12.1 | 5.3  | 0.5                   | 130  | 0.618 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 19.2                   | 83.3            | 64.0 | 80.2 | 62.6 | 77.2 | 61.4 | 66.8                | 78.0 | 65.2 | 76.0 | 63.6 | 73.3 | 6.2                  | 140  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 63.2                            | 92.3 | 70.3 | 61.4 | 86.7 | 69.0 | 59.8 | 81.6 | 68.0 | 32.5          | 78.9 | 31.2 | 76.0 | 29.9 | 73.6 | 76.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 13.5          | 11.6 | 9.9 | DB                         | -35.8 | 89.1               | 8.3 | 3.6                                                | -41.7 | 91.7       | -46.6 | 93.8       | -51.2 | 95.8       | -57.3 | 98.4 | (4) |
| (5)               |      |     | WB                         | -35.9 | 69.8               | 8.3 | 2.4                                                | -41.8 | 71.5       | -46.6 | 72.9       | -51.3 | 74.3       | -57.3 | 76.0 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 32.3   | -2.3  | 4.3   | 19.4  | 35.9  | 48.2  | 60.0  | 64.5  | 60.0  | 47.4  | 32.9  | 14.1  | 1.1   |  |  |
|                                                                        | DBStd       | 25.44  | 12.63 | 12.48 | 11.56 | 9.06  | 9.10  | 7.29  | 5.28  | 6.46  | 8.04  | 9.91  | 12.93 | 13.29 |  |  |
|                                                                        | HDD50       | 7712   | 1621  | 1281  | 950   | 430   | 147   | 9     | 0     | 6     | 143   | 532   | 1077  | 1516  |  |  |
|                                                                        | HDD65       | 12064  | 2086  | 1701  | 1415  | 872   | 524   | 181   | 74    | 179   | 529   | 995   | 1527  | 1981  |  |  |
|                                                                        | CDD50       | 1241   | 0     | 0     | 0     | 8     | 92    | 308   | 449   | 317   | 65    | 2     | 0     | 0     |  |  |
|                                                                        | CDD65       | 119    | 0     | 0     | 0     | 0     | 4     | 31    | 58    | 25    | 1     | 0     | 0     | 0     |  |  |
|                                                                        | CDH74       | 1405   | 0     | 0     | 0     | 2     | 94    | 423   | 602   | 263   | 21    | 0     | 0     | 0     |  |  |
|                                                                        | CDH80       | 287    | 0     | 0     | 0     | 0     | 16    | 92    | 127   | 50    | 2     | 0     | 0     | 0     |  |  |
|                                                                        | WSAvg       | 4.0    | 2.7   | 3.2   | 4.3   | 5.2   | 5.4   | 4.5   | 4.0   | 4.0   | 4.1   | 4.1   | 3.7   | 2.9   |  |  |
|                                                                        | PrecAvg     | 16.4   | 0.5   | 0.4   | 0.3   | 0.8   | 1.3   | 2.2   | 3.5   | 3.5   | 1.7   | 0.9   | 0.7   | 0.7   |  |  |
| (15) Precipitation                                                     | PrecMax     | 22.6   | 1.6   | 1.0   | 1.0   | 2.3   | 3.0   | 5.2   | 7.2   | 8.3   | 3.3   | 3.0   | 1.2   | 1.5   |  |  |
|                                                                        | PrecMin     | 10.7   | 0.1   | 0.2   | 0.0   | 0.1   | 0.5   | 0.9   | 0.6   | 1.4   | 0.6   | 0.0   | 0.1   | 0.2   |  |  |
|                                                                        | PrecStd     | 2.8    | 0.3   | 0.2   | 0.2   | 0.6   | 0.6   | 0.9   | 1.5   | 1.7   | 0.7   | 0.6   | 0.3   | 0.3   |  |  |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 29.3  | 35.2  | 47.4  | 68.2  | 82.5  | 86.8  | 88.0  | 85.2  | 77.1  | 62.8  | 43.7  | 32.2  |  |  |
|                                                                        |             | MCWB   | 26.4  | 28.1  | 36.7  | 48.8  | 60.0  | 64.2  | 67.3  | 64.7  | 59.1  | 47.5  | 36.1  | 27.2  |  |  |
|                                                                        | 2%          | DB     | 24.8  | 29.6  | 42.7  | 61.2  | 76.5  | 82.6  | 83.4  | 80.4  | 70.3  | 56.0  | 38.4  | 25.9  |  |  |
|                                                                        |             | MCWB   | 22.7  | 25.1  | 34.4  | 45.8  | 55.2  | 62.5  | 65.5  | 63.7  | 55.7  | 44.5  | 33.1  | 23.2  |  |  |
|                                                                        | 5%          | DB     | 19.4  | 25.7  | 39.0  | 55.8  | 71.0  | 79.0  | 80.2  | 76.6  | 65.3  | 50.9  | 35.0  | 22.2  |  |  |
|                                                                        |             | MCWB   | 17.4  | 22.0  | 32.0  | 42.8  | 52.5  | 60.9  | 64.3  | 61.9  | 53.3  | 41.9  | 31.0  | 20.2  |  |  |
|                                                                        | 10%         | DB     | 14.5  | 21.3  | 35.6  | 50.9  | 65.6  | 75.3  | 77.2  | 72.8  | 60.6  | 46.4  | 31.4  | 18.2  |  |  |
|                                                                        |             | MCWB   | 13.1  | 18.6  | 29.7  | 40.4  | 49.7  | 59.4  | 62.9  | 60.4  | 51.2  | 39.6  | 28.5  | 16.8  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 27.0  | 29.3  | 37.8  | 50.0  | 60.6  | 67.4  | 70.3  | 67.3  | 61.3  | 49.4  | 37.3  | 27.2  |  |  |
|                                                                        |             | MCDB   | 28.2  | 33.8  | 45.6  | 63.5  | 79.4  | 78.7  | 81.7  | 79.9  | 74.8  | 59.4  | 42.5  | 31.1  |  |  |
|                                                                        | 2%          | WB     | 22.5  | 25.4  | 34.9  | 47.0  | 56.8  | 65.2  | 67.6  | 65.0  | 57.2  | 45.7  | 33.6  | 23.6  |  |  |
|                                                                        |             | MCDB   | 24.6  | 29.4  | 42.0  | 59.3  | 73.0  | 77.6  | 78.0  | 75.9  | 67.0  | 54.0  | 37.5  | 25.9  |  |  |
|                                                                        | 5%          | WB     | 17.5  | 22.1  | 32.4  | 43.8  | 53.9  | 63.2  | 66.2  | 63.4  | 54.6  | 42.9  | 31.3  | 20.2  |  |  |
|                                                                        |             | MCDB   | 19.4  | 25.5  | 38.3  | 54.9  | 68.2  | 74.7  | 77.4  | 73.6  | 63.2  | 50.6  | 34.5  | 22.1  |  |  |
|                                                                        | 10%         | WB     | 13.1  | 18.6  | 29.9  | 40.6  | 51.0  | 61.1  | 64.7  | 61.6  | 52.1  | 40.0  | 28.6  | 16.8  |  |  |
|                                                                        |             | MCDB   | 14.6  | 21.3  | 35.4  | 50.5  | 64.0  | 72.3  | 74.2  | 70.7  | 59.1  | 45.6  | 31.6  | 18.2  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 14.0  | 18.3  | 19.3  | 17.7  | 21.5  | 21.2  | 19.2  | 18.4  | 18.0  | 15.7  | 13.1  | 12.6  |  |  |
|                                                                        |             | MCDBR  | 14.2  | 19.9  | 21.2  | 25.9  | 29.8  | 27.9  | 25.5  | 25.6  | 26.0  | 23.6  | 14.8  | 14.4  |  |  |
|                                                                        |             | MCWBR  | 12.7  | 16.1  | 14.6  | 14.4  | 14.3  | 11.4  | 10.6  | 10.6  | 12.4  | 13.2  | 11.0  | 12.5  |  |  |
|                                                                        |             | MCDBR  | 13.9  | 19.0  | 19.0  | 24.4  | 27.1  | 23.2  | 21.3  | 22.0  | 22.9  | 21.5  | 14.0  | 13.9  |  |  |
|                                                                        | 5% WB       | MCWBR  | 12.6  | 15.5  | 13.3  | 14.5  | 13.7  | 10.5  | 10.1  | 10.2  | 11.6  | 12.4  | 10.9  | 12.3  |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.273 | 0.304 | 0.337 | 0.377 | 0.431 | 0.419 | 0.417 | 0.388 | 0.344 | 0.328 | 0.299 | 0.274 |  |  |
|                                                                        | taud        |        | 2.047 | 2.029 | 2.039 | 2.192 | 2.096 | 2.241 | 2.275 | 2.339 | 2.420 | 2.326 | 2.152 | 2.006 |  |  |
|                                                                        | Ebn at Noon |        | 203   | 239   | 260   | 268   | 258   | 263   | 261   | 262   | 261   | 234   | 192   | 169   |  |  |
|                                                                        | Edh at Noon |        | 22    | 32    | 40    | 40    | 47    | 41    | 39    | 35    | 28    | 24    | 20    | 18    |  |  |
|                                                                        | RadAvg      |        | 322   | 658   | 1176  | 1532  | 1701  | 1893  | 1714  | 1435  | 1000  | 582   | 306   | 218   |  |  |
| (45) All-Sky Solar Radiation                                           | RadStd      |        | 19    | 31    | 49    | 116   | 150   | 108   | 147   | 92    | 92    | 47    | 30    | 21    |  |  |

## Historical Trends

|                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                        |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (8 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page